

One platform. Every science lab. Powered by AI.

Free, browser-based chemistry, physics, and biology simulations guided by an AI tutor that knows exactly what you are exploring.

PURPOSE

Hundreds of millions of students attend schools without functioning science laboratories. They memorise reactions they have never seen, calculate trajectories they have never traced, and label cell diagrams they have never found under a microscope. SciSim AI removes this barrier entirely — replacing physical equipment with real-time interactive simulations guided by a context-aware AI tutor, at zero cost, in any modern browser.

THE THREE LABS

Chemistry Lab

- Mix elements from a virtual periodic table
 - Reactions animate live — colour shift, rising bubbles, temperature gauge
 - Automatic safety warnings for hazardous products
 - 3 presets: neutralisation, combustion, electrolysis
- e.g. $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
 $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$

Physics Lab

- Live sliders update simulations in real time
 - Pendulum governed by $\alpha = -\frac{g}{L} \sin \theta$
 - Projectile arc: $x = v_0 \cos \theta \cdot t$, $y = v_0 \sin \theta \cdot t - \frac{1}{2}gt^2$
 - Wave interference computed pixel-by-pixel:
 $d = A \sin(kr_1 - \omega t) + A \sin(kr_2 - \omega t)$
- Range, max height, period displayed live.

Biology Lab

- Interactive 3D cell models built in Three.js
 - Click any organelle to highlight and learn its function
 - Animal cell, plant cell, and mitosis specimens
 - Raycasting for precise click detection at any zoom
- Nucleus, mitochondria, ER, chloroplasts, vacuole, ribosomes — all interactive.

AI Lab Partner

Powered by **Claude** (claude-sonnet-4-6). Knows the active experiment, live variable values, and recent observations. Answers questions in plain language, suggests next steps, and flags real-world applications.

Lab Report Generator

One click after any session produces a complete structured report: *objective, hypothesis, materials, procedure, observations, results, analysis, conclusion*, and safety notes — ready to download.

Text-to-Experiment

Describe what you want to explore in plain English. The AI maps the description to a simulation type, selects appropriate variable values, and launches the experiment automatically — no manual configuration needed.

TECH STACK

Frontend	React 18, Vite, Tailwind CSS, Framer Motion
Simulations	p5.js (Chem & Physics), Three.js + R3F (Bio)
AI Layer	Anthropic Claude API (claude-sonnet-4-6)
Routing	React Router v6
Charts	Recharts
Deploy	Vercel (zero-config, instant CDN)

IMPACT & ACCESSIBILITY

SciSim requires **no login, no installation, and no budget** — just a browser. It is optimised for low-bandwidth connections and works on the affordable smartphones that are the primary computing device for students across South Asia, Sub-Saharan Africa, and Latin America. Science education must not depend on geography or school funding.